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Modulo Operation
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Monoalphabetic Cipher
Natural Language Redundancy
Polyalphabetic Cipher
Pseudo-Code: Vigenere Cipher

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Modulo Operation and Caesar Cipher

Review Learning Objectives

1. Let's use a modified Caesar Cipher where $c = (p + 2x) \bmod 26$, where c and p are the ciphertext and the plaintext, respectively, and x is the key. How many distinct keys, producing distinct encryption/decryption transformations, are there now?

4 / 4 points

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Due Apr 7, 11:59 PM ISTAttempts 3 every 8 hours

Correct

See Modulo Operation lesson and interpret the equation.

Requestion

To Pass 65% or higher

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2. For $c = (p + 2x) \bmod 26$, where c , p , and x are the ciphertext, the plaintext, and the key, respectively, what is the corresponding decryption? Select all that works.

4 / 4 points

☐ $p = (c - x) \bmod 26$

☒ $p = (c + 26 - 2x) \bmod 26$

Correct

See Modulo Operation

☐ $p = (c - x) \bmod 13$

☒ $p = (c - 2x) \bmod 26$

Correct

See Modulo Operation lesson.

☐ $p = (c - 2x) \bmod 13$

3. Let's now use a modified Caesar Cipher where $c = (p + 3x) \bmod 26$, where c and p are the ciphertext and the plaintext, respectively, and x is the key. How many distinct keys, producing distinct encryption/decryption transformations, are there now?

4 / 4 points

26

Correct

See Modulo Operation lesson and interpret the equation.

https://www.coursera.org/learn/symmetric-crypto/exam/vBd0I/modulo-operation-and-caesar-cipher/attempt/true

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